Learning Objectives

- To learn the process of fuel filling in self-propelled vehicles.
- To learn about important components like carburettor, pump, etc.

9.0 INTRODUCTION

In this chapter we study about the Fuel injection system (types, parts)-Air fuel ratio-Carburettor- types (simple, solex) and Types of Diesel engines pump, Types of Governors, Advance mechanisms, Reasons for knocking (measures to control knocking), Injector and Different types of fuel supply system i.e DTSI, CCTI, VT, PGMFI, MPFI. Fig 9.0 shows the general layout of Fuel Supply System.

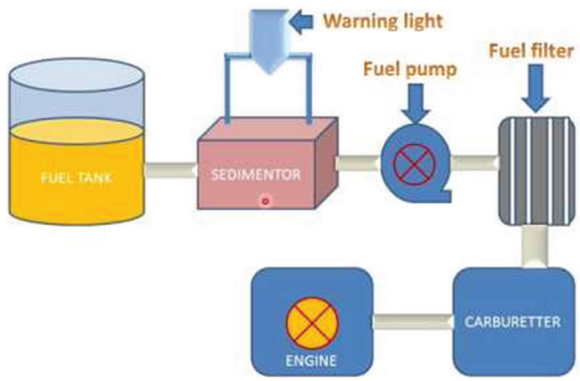


Figure 9.0 Fuel Supply System

9.1 FUEL SUPPLY SYSTEM IN PETROL ENGINE

The petrol from the petrol tank and the air from the air filter is mixed completely in correct ratio in order to make the fuel burn completely (stoichiometric air fuel ratio). The mixed fuel is sent in to the engine cylinder continuously according to the speed, load and torque of the engine in exact pressure. Fig 9.0 shows the general layout of Fuel Supply System in Petrol Engine.

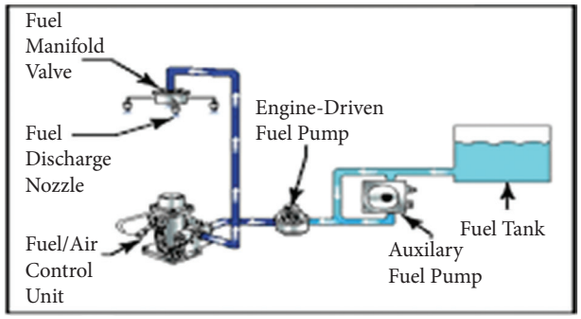


Figure 9.1 Fuel Supply System in Petrol Engine

9.2 TYPES OF FUEL SUPPLY SYSTEM

1. Gravity system
2. Vacuum feed system
3. Pump system
4. Injection system

9.2.1 Gravity System

In this method the petrol tank is kept just above the engine setup. So that the petrol in the tank reaches the Carburettor due to the gravity force. The air from the air filter and the petrol from the tank mixes and supplied in to the inlet manifold of the engine. This methods are commonly used in two wheelers like moped, scooter and motorcycles. This method can't be used in multi cylinder and heavy duty engines because as it has to be placed near the

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